Name: Sagi Sahasra Srihasa
Roll: 2520030008

Step1:

```sql
/*
Name: Sagi Sahasra Srihasa
Roll: 2520030008
*/
CREATE DATABASE BankDB2;
USE BankDB2;

CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);

CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```

✔	23	12:09:50	CREATE DATABASE BankDB2	1 row(s) affected
✔	24	12:09:52	USE BankDB2	0 row(s) affected
✔	25	12:09:54	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT N...	0 row(s) affected
✔	26	12:09:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC...	0 row(s) affected

Step 2:

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (Account_No) REFERENCES Account(Account_No)
);

CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```

✔	27	12:10:29	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account...	0 row(s)
✔	28	12:10:31	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s)

Step 3:

```sql
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Sahasra', '9876543210', 'sahasra@gmail.com', 'Hyderabad'),
(102, 'Kriti Sharma', '9876543211', 'kriti@gmail.com', 'Vijayawada'),
(103, 'Arjun', '9876543212', 'arjun@gmail.com', 'Bangalore'),
(104, 'Snehitha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
(105, 'Kiran ', '9876543214', 'kiran@gmail.com', 'Hyderabad'),
(106, 'Anil Kumar', '9876543215', 'anil@gmail.com', 'Delhi'),
(107, 'Meena Reddy', '9876543216', 'meena@gmail.com', 'Mumbai'),
(108, 'Rahul Sharma', '9876543217', 'rahul@gmail.com', 'Pune'),
(109, 'Lakshmi Devi', '9876543218', 'lakshmi@gmail.com', 'Hyderabad'),
(110, 'Suresh Babu', '9876543219', 'suresh@gmail.com', 'Vijayawada');
```

29 12:10:53 INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Sahasra', '987

Step 4:

```sql
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad'),
(10006, 106, 'Savings', 65000, 'Delhi'),
(10007, 107, 'Current', 150000, 'Mumbai'),
(10008, 108, 'Savings', 35000, 'Pune'),
(10009, 109, 'Savings', 85000, 'Hyderabad'),
(10010, 110, 'Current', 110000, 'Vijayawada');
```

30 12:11:57 INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '

Step 5:

```sql
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

31 12:12:23 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 1

Step 6:

```sql
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5),
(505, 105, 'Home Loan', 4000000, 7.2),
(506, 106, 'Car Loan', 900000, 8.5),
(507, 107, 'Business Loan', 3000000, 9.0),
(508, 109, 'Personal Loan', 600000, 10.0);
```

32 12:26:38 INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, '...

Step 7:

```sql
SELECT * FROM Customer;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Sahasra	9876543210	sahasra@gmail.com	Hyderabad
102	Kriti Sharma	9876543211	kriti@gmail.com	Vijayawada
103	Arjun	9876543212	arjun@gmail.com	Bangalore
104	Snehitha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran	9876543214	kiran@gmail.com	Hyderabad

Step 8:

```sql
SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Step 9:

```sql
SELECT * FROM Bank_Transaction;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 12:12:23
2	10001	WITHDRAW	5000.00	2026-08-15 12:12:23
3	10002	DEPOSIT	15000.00	2026-08-15 12:12:23
4	10003	WITHDRAW	20000.00	2026-08-15 12:12:23
5	10004	DEPOSIT	5000.00	2026-08-15 12:12:23

Step 10:

```sql
SELECT * FROM Loan;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20

Step 11:

```sql
CREATE VIEW Customer_View AS
SELECT *
FROM Customer;

SELECT * FROM Customer_View;
```

37 12:28:19 CREATE VIEW Customer_View AS SELECT * FROM Customer

Step 12:

```sql
SELECT * FROM Customer_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Sahasra	9876543210	sahasra@gmail.com	Hyderabad
102	Kriti Sharma	9876543211	kriti@gmail.com	Vijayawada
103	Arjun	9876543212	arjun@gmail.com	Bangalore
104	Snehitha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran	9876543214	kiran@gmail.com	Hyderabad

Step 13:

```sql
CREATE VIEW Savings_Account_View AS
SELECT *
FROM Account
WHERE Account_Type = 'Savings';
```

39 12:29:12 CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_

Step 14:

```sql
SELECT * FROM Savings_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune

Step 15:

```
CREATE VIEW Customer_Account_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID;
```

41 12:37:55 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name

Step 16:

```
SELECT * FROM Customer_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Sahasra	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Kriti Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Arjun	Bangalore	10003	Current	120000.00	Bangalore
104	Snehitha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran	Hyderabad	10005	Current	90000.00	Hyderabad

Step 17:

```
CREATE VIEW Total_Bank_Balance AS
SELECT
    SUM(Balance) AS Total_Balance
FROM Account;
```

43 12:38:41 CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account

Step 18:

```sql
SELECT * FROM Total_Bank_Balance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

Total_Balance
825000.00

Step 19:

```sql
CREATE VIEW Multiple_Account_Branches AS
SELECT
    Branch,
    COUNT(*) AS Number_of_Accounts
FROM Account
GROUP BY Branch
HAVING COUNT(*) > 1;
```

45 12:39:30 CREATE VIEW Multiple_Account_Branches AS SELECT Branch, COUNT(*) AS Number_of_Accounts F...

Step 20:

```sql
SELECT * FROM Multiple_Account_Branches;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2

Step 21:

```sql
CREATE VIEW Loan_Interest_View AS
SELECT
    Loan_ID,
    Customer_ID,
    Loan_Type,
    Loan_Amount,
    Interest_Rate,
    (Loan_Amount * Interest_Rate / 100) AS Annual_Interest
FROM Loan;
```

47 12:40:04 CREATE VIEW Loan_Interest_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_A

Step 22:

```sql
168 • SELECT * FROM Loan_Interest_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
504	104	Personal Loan	500000.00	10.50	52500.00000000
505	105	Home Loan	4000000.00	7.20	288000.00000000
506	106	Car Loan	900000.00	8.50	76500.00000000
507	107	Business Loan	3000000.00	9.00	270000.00000000
508	109	Personal Loan	600000.00	10.00	60000.00000000